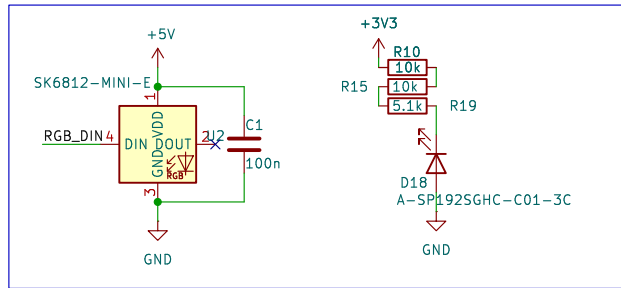
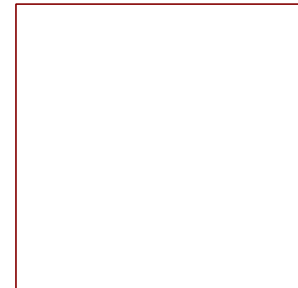


LEDs

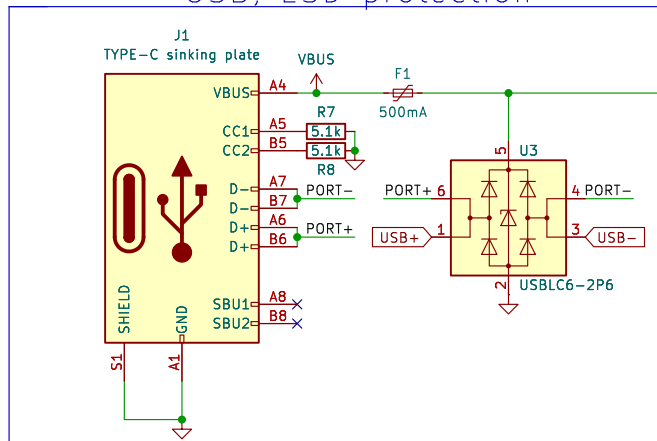


Switch Matrix

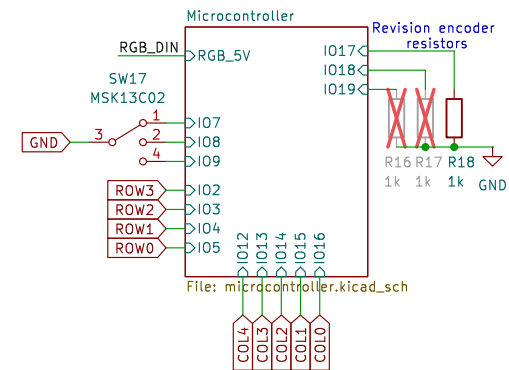
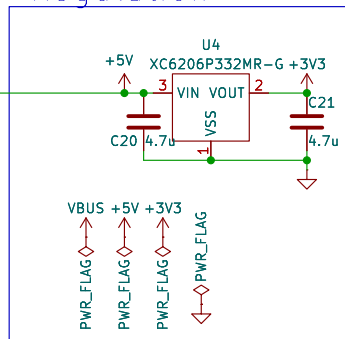


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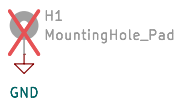
USB, ESD protection



Regulation



File: microcontroller.kicad_sch



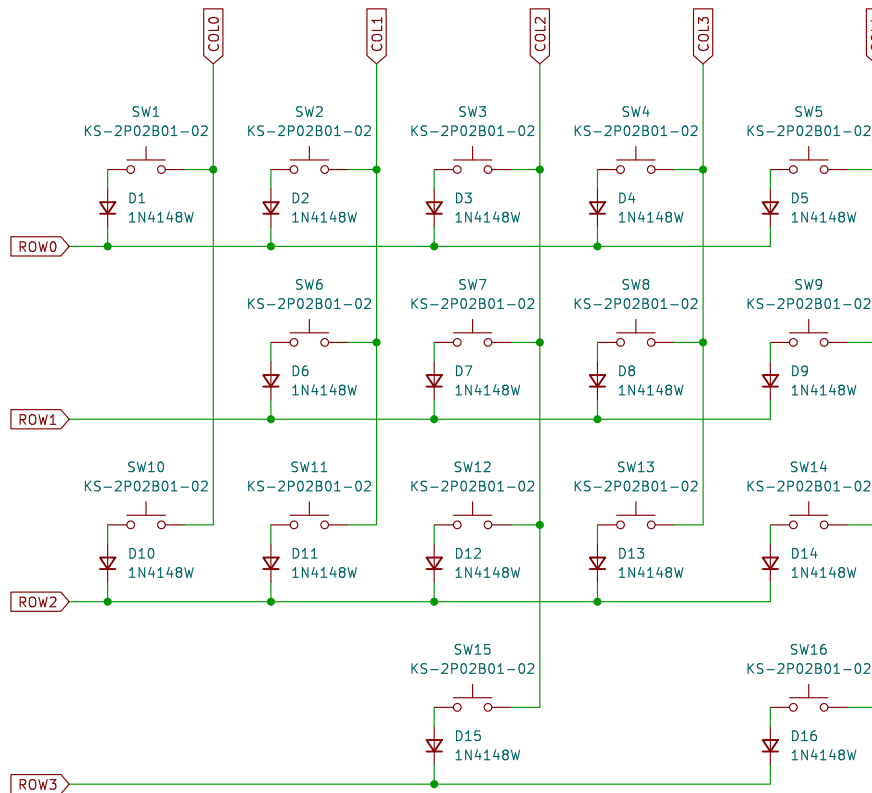
Summacogni OÜ

Sheet: /
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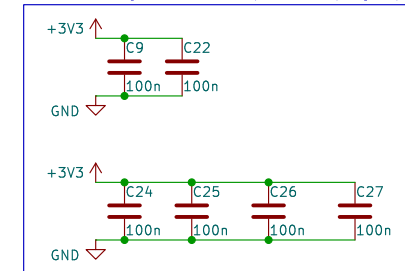
Title: Mathpad PCB

Size: A4 Date: 2025-04-07
KiCad E.D.A. 8.0.4

Rev: 1.7
Id: 1/3



Matrix scanning current return path decoupling caps



Summacogni OÜ

Sheet: /Switch Matrix/
File: switch_matrix.kicad_sch

Title: Mathpad PCB

Size: A4 Date: 2025-04-07

KiCad E.D.A. 8.0.4

Rev: 1.7

Id: 2/3

The schematic diagram illustrates the RP2040 microcontroller board. The RP2040 chip is the central component, with its pins connected to various external components. The components include:

- USB Connectors:** USB+, USB-, D+, and D- pins are connected to the RP2040.
- Crystal and Capacitors:** A 12MHz crystal (Y1) is connected to XTAL_IN and XTAL_OUT pins, with 30pF capacitors (C6, C8) and a 1k resistor (R14).
- Resistors:** A 10k resistor (R11) is connected to the SWCLK pin, and a 1k resistor (R14) is connected to the XTAL_OUT pin.
- Capacitors:** A 100nF capacitor (C18) is connected to the +1V1 pin.
- Level Shifter:** A 74LVC1T45 level shifter (U7) is used to interface the 3.3V logic of the RP2040 with a 5V USB power source. The VCCA pin is connected to +3V3, the VCCB pin is connected to +5V, and the DIR pin is connected to +3V3.
- Power Supply:** The RP2040 is powered by a +3V3 source, which is connected to the VREG_VIN pin.

The RP2040 pinout is listed on the right side of the chip:

Pin	Function
2	0_TX0
3	1_RX0
4	2_SDA1
5	3_SCL1
6	4
7	5
8	6
9	7
10	8
11	9
12	10
13	11
14	12
15	13
16	14
17	15
18	16
19	17
20	18
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99	97
100	98

Id: 3/3